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RESEARCH ARTICLE

A Scalable AI Approach to Bird Species Identification for Conservation and Ecological Planning

SOUAD HASSANIE^{1,2}, AIMAN GOHAR³, RAJA HASHIM ALI^{2,3}, TALHA ALI KHAN²,
IFTIKHAR AHMED², AND SHAN FAIZ²

¹Business School, Holy Spirit University of Kaslik, Jounieh, Lebanon

²Department of Business, University of Europe for Applied Sciences, Think Campus, 14469 Potsdam, Germany

³Department of Artificial Intelligence, Faculty of Computer Science and Engineering, GIK Institute of Engineering Sciences and Technology, Topi 23460, Pakistan

Corresponding author: Raja Hashim Ali (hashim.ali@ue-germany.de)

ABSTRACT Birds represent an important component of ecosystems, holding substantial ecological and human interest, and significantly contributing to ecological equilibrium via their roles as pollinators, seed dispensers, and predators. If identified correctly, the wild life departments, farmers, ecologists, and various other experts can ensure saving endangered bird species, improving crop yield and ensure seed safety, and plan safe sanctuaries for birds, etc. Due to the vast array of bird species and the often subtle distinctions among them, human-driven manual classification poses a formidable challenge. The manual classification of bird species is a labor-intensive and error-prone process, predominantly reliant on domain experts. However recent algorithmic developments and progress in the deep-learning based object detection and classification models has made this difficult and time-consuming task easy and accurate. Nevertheless, numerous existing CNN models have been trained or fine-tuned on limited datasets, rendering them ill-equipped to handle the classification of a substantial volume of diverse bird species. To address this challenge, this study introduces a novel approach based on transfer learning and the ResNet-50 model to ensure accurate identification of endangered species. Our methodology involved fine-tuning the ResNet-50 model on the 100 Bird Species dataset, with the training data consisting of 85,000 images, representing 525 distinct bird species with each species comprising of 130 or more diverse pre-processed images, including 25 endangered and 12 critically endangered species. The transfer learning based mechanism applied on the large dataset achieved a state-of-the-art accuracy of 87% in bird species classification through deep learning. Our results showed that out of 12 critically endangered species like *California condor*, *Ashy storm petrel*, *California quail* and other quails, none is misclassified by our system. Similarly, none of the 25 endangered species is misclassified by our system, indicating a solid performance for use in identification of endangered and critically endangered bird species. This study shows that deep learning and transfer learning can be successfully deployed to identify birds, and ensure the safety of endangered and critically endangered species of birds, making it a powerful and adaptable approach for the future.

INDEX TERMS Bird species classification, convolutional neural networks, deep learning, endangered species, machine learning, transfer learning.

I. INTRODUCTION

Classification and correct identification of birds is pivotal for comprehending avian diversity, ecology, and behavior, and is essential for a broad spectrum of applications,

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including conservation, ecological research, agriculture, and public health management. Through precise identification and categorization of bird species, researchers and practitioners can make informed decisions to advance biodiversity conservation, ecosystem health, and human welfare [1]. The deployment of deep learning and transfer learning techniques provides a robust mechanism for automated bird species identification. Such a mechanism can help in the identification and monitoring of bird species across diverse ecosystems, enabling conservationists to evaluate ecosystem health, pinpoint habitats in need of protection, and monitor the impact of human activities on bird populations [2]. These insights are indispensable for crafting effective conservation strategies and policies geared towards preserving biodiversity [3]. Birds assume critical roles in ecosystems as pollinators, seed dispersers, and predators, underscoring the importance of studying their populations and interactions with other species to unravel ecosystem dynamics. By categorizing bird species based on their ecological roles, behaviors, and/or habitat preferences, the mechanism facilitates research on topics such as species interactions, community ecology, and migration patterns [4]. Birds, renowned for their responsiveness to climatic fluctuations, ecological shifts, and habitat degradation, can serve as interesting indicators of environmental health [5]. Additionally, birds influence agricultural systems both positively and negatively, serving as natural pest controllers or causing crop damage [6]. Consequently, accurate bird identification will help farmers in distinguishing between beneficial and harmful bird species, empowering them to implement tailored management strategies for crop protection and habitat conservation. Thus, accurate bird identification is critical and extremely useful in a variety of fields and scientific endeavors.

Bird classification from images using computer vision techniques has gained popularity in recent times but poses a challenge, especially when dealing with a over 10,000 species, noisy and non-standardized images, image labeling, or related issues [7]. Due to these complexities, deep learning based models have exhibited marked enhancements over traditional computer vision approaches for object classification, in particular for avian species classification [8], [9], [10]. Presently, deep learning models have become the standard choice for classifying images, finding applications in various domains such as medical image analysis, traffic management, autonomous driving, and robotic vision. Following the success of AlexNet and similar deep learning architectures, there has been a surge in research papers employing deep learning models for object classification tasks [11]. Nevertheless, these models heavily rely on the availability of extensive training data, with data availability and data labeling remaining significant challenges for object detection and species identification. Consequently, techniques like transfer learning, which focus on training only a subset of the model's layers, have proven successful, particularly in scenarios with limited sample sizes [10]. Additionally, access to large, publicly available datasets has

enhanced the accuracy of deep learning models for better classification performance in these application domains [11]. The deployment of deep learning techniques not only enables us to fulfill our mission of biodiversity conservation but also serves as a testament to how technological innovation can effectively address pressing environmental challenges.

Multiple studies have addressed the use of deep learning for bird species classification in recent years. Numerous deep learning techniques, such as VGG-16 [12], MobileNetV2 [10], and DCNN [13], have been used for bird classification. Transfer learning techniques have demonstrated notable success in this domain, as evidenced from the work of Choudhary et al. [14], who utilized a transfer learning approach on a 100 Bird Species dataset containing 200 distinct bird species, yielding superior results. Similarly, Madhavi et al. [15] deployed pre-trained VGG-16 and ResNet-50 models to achieve accurate results. Ragib et al. [16] employed a pre-trained ResNet model and combined it with a base model to encode images and achieved higher accuracy than earlier models. However, none of these studies has focused specifically on identification of endangered bird species.

The aim of this study is to address the task of bird species identification from images in general, and identification of endangered species of birds in particular. The approach taken involves the utilization of a pre-trained Resnet-50 [17] model, fine-tuned on an extensive dataset [18] which comprises of over 90,000 high-quality images with 525 distinct bird species out of which 37 are either endangered or critically endangered. The results from the current study are useful for research as well as have implications for ecological conservation [19] and the responsible management of birds in the ecosystem. The special focus is to improve the identification of endangered species using state-of-the-art deep learning models and this research is our contribution to facilitating conservation of endangered bird species, ultimately facilitating the sustainable coexistence of humans and birds in our shared ecosystem. The current study builds upon the foundations laid by prior research endeavors and intends to improve accuracy and efficiency in endangered bird species classification, using ResNet-50 model.

II. LITERATURE REVIEW

The application of deep learning techniques in bird species classification has witnessed substantial growth over the past decade. In recent years, the field of bird species recognition has experienced remarkable advancements due to the application of deep learning techniques. A well-established approach for bird species classification, namely transfer learning, has consistently demonstrated exceptional performance in several studies. In recent times, this approach has been used in several studies, including Kumar and Das [20], Wu et al. [21], Choudhary et al. [14], Madhavi et al. [15], Kondaveeti et al. [10], Gupta et al. [22], and Ragib et al. [16]. Despite challenges such as issues like bird posture variations and complex background, pioneering techniques proposed in

the works of Huang and Basanta [23], Liu et al. [24], and Li et al. [25] have underscored the significance of this task in the domains of image classification, multimedia processing, and computer vision. The scope of bird classification extends beyond the image datasets as many researches and work has shown remarkable results in birds classification task using audio recordings, as shown by Rajan and Noumida [12] and Noumida et al. [26]. Zhang et al. [27] introduced a model generalisability technique, while Gavali and Banu [13] employed GoogleNet framework and parallel processing to achieve high performance in the domain of bird species classification, where the authors tackled the challenges of bird species identification by employing deep convolutional neural networks (DCNN) on a GPU platform with a prediction accuracy of 88.33%.

Transfer learning techniques have yielded exceptional results in the task of classifying bird species, drawing significant attention from various researchers. Choudhary et al. [14] employed a transfer learning approach, utilizing a dataset featuring 200 distinct bird species. Their findings strongly support the usefulness of transfer learning in achieving superior outcomes in this domain. Madhavi et al. [15] harnessed pre-trained models, specifically VGG-16 and Resnet-50, which led to remarkable results in their bird species classification endeavors. Kumar and Das [20] adopted a pre-trained Region-based Convolutional Neural Network (R-CNN) and assembled an ensemble model that incorporated Inception Nets, including InceptionV3 net and InceptionResnetV2, allowing them to simultaneously determine the localization and species of birds in images. Meanwhile, Wu et al. [21] leveraged deep transfer learning for bird classification, using the CUB-200-2011 dataset and ResNeXt for transfer learning. Kondaveeti et al. [10] applied a deep learning transfer approach, utilizing the lightweight architecture of MobileNetV2, with an accuracy rate of 96.79% utilizing two distinct datasets (first dataset of 200 bird species with 1800 images of a single bird, and second dataset of 20 native South India bird species with 180 images of a single bird) to train MobileNetV2. Gupta et al. [22] developed a mobile application capable of classifying a diverse array of 260 bird species through a transfer learning approach using convolutional neural networks (CNNs) pre-trained on the ImageNet Dataset, encompassing models such as EfficientNet-lite0, Xception, MobileNetV2, ResNet-50, InceptionV3, and InceptionResNetV2. These models were fine-tuned on a dataset comprising images of 260 different bird species, achieving a maximum accuracy of 99.82% on the training data and an impressive 98.61% on the testing data. Ragib et al. [16] employed a pre-trained ResNet model and combined it with a base model to encode images with the primary objective of identification of birds of varying sizes, shapes, and colors in input images, with an accuracy of 97.98%. In another study, Potluri et al. [28] used a pre-trained ResNet model, further fine-tuning it on the Caltech-UCSD Birds 200 dataset, with an accuracy of 96.5%.

Due to interclass similarities and the presence of fine-grained features, Huang and Basanta [23] devised a transfer learning approach based on fivefold cross-validation (with misclassified data points exchanged between the training and validation datasets) and Inception-ResNet-v2 to identify and classify bird species that are indigenous to Taiwan. When tested on 760 images representing various bird species native to Taiwan, the model achieved an impressive accuracy rate of 98.39%. On the other hand, Liu et al. [24] introduced the Invariant Cues-Aware Feature Connection Transformer (TransIFC) to capture semantic relationships within bird images incorporating two innovative modules: the Hierarchy Stage Feature Aggregation (HSFA) module, which aggregates multiscale information in bird images by concatenating features from various layers, and the Feature in Feature Abstraction (FFA) module, which employs feature selection based on discrimination errors to extract invariant cues specific to birds. Their model achieved accuracy of 91.0% on the CUB-200-2011 and 90.9% on the NABirds datasets. In Yunnan province, China, which boasts a total of 1074 bird species and subspecies, Li et al. [25] created their own dataset known as YUB-300-2019 for bird images and a ResNet and a joint semantic components technique based model for the fine-grained recognition of these bird images. The method demonstrated a high level of accuracy, suggesting that it has the potential for broader applications in bird identification and classification.

The classification of bird species from audio recordings is also an interesting mechanism for bird species identification. Rajan and Noumida [12] employed a transfer learning approach to identify multiple bird species within the Xeno-canto bird sound database, utilizing a pre-trained VGG-16 network with an average F1-score of 0.65, surpassing the performance of the acoustic cue-based MFCC-DNN approach. In another study, Noumida et al. [26] conducted a detailed analysis of melspectrograms using a sliding window technique to learn the vocalizations of birds and identify multiple bird species from audio recordings. Their method was based on ensemble fusion models and a subsequent sequential aggregation strategy, applied to the Xeno-canto bird sound database with an F1-Score of 0.60.

In order to enhance the model's capacity to generalize, Zhang et al. [27] created an extensive dataset featuring 275 bird species in various environmental conditions, including rainy and foggy settings. To streamline the model's complexity, they introduced a lightweight neural network called ShuffleNetV2 and incorporated a multi-channel feature fusion structure (MCF) to enhance the network's adaptability for extracting information across multiple-channel scales. The outcomes revealed that their proposed approach achieved an accuracy rate of 92.3%, surpassing the accuracy of ShuffleNetV1 (86.2%) and ShuffleNetV2 (90.5%).

A table summarizing the characteristics of the existing literature along with the novelty of the proposed work is shown (as shown in Table 1).

TABLE 1. Literature review table showing the contributions of various authors for Birds' species classification.

Study	Year of publication	Dataset Used	Based on images	Resolution greater than 400*400	More than 80,000 images	More than 500 species	Identifies Endangered species	ResNet-50	Transfer learning	Accuracy reported for endangered species
Kumar et al.	2019	CVIP 2018 Bird Species challenge	✓	✓					✓	
Ragib et al.	2020	CUB-200-2011	✓					✓	✓	
Choudhary et al.	2021	100 Bird Species v1	✓	✓					✓	
Wu et al.	2021	CUB-200-2011	✓					✓	✓	
Gupta et al.	2021	CUB-200-2011	✓					✓	✓	
Huang et al.	2021	Internet of Birds	✓	✓			✓	✓	✓	
Rajan et al.	2021	Xeno-canto bird sound	×						✓	
Li et al.	2022	YUB-300-2019	✓					✓	✓	
Madhavi et al.	2023	Internet of Birds	✓	✓				✓	✓	
Kondaveeti et al.	2023	CUB-200-2011 + A self-made dataset	✓						✓	
Liu et al.	2023	CUB-200-2011 + NABirds	✓							
Noumida et al.	2023	Xeno-canto bird sound	×							
Potluri et al.	2023	CUB-200-2011	✓					✓	✓	
Zhang et al.	2023	A self-made dataset	✓	✓						
Proposed work	2024	100 Bird Species v2	✓	✓	✓	✓	✓	✓	✓	✓

III. OUR CONTRIBUTION

A. GAP ANALYSIS

Following are a few gaps in the literature that we found while conducting this study.

- 1) **Endangered species:** While the bird species identification using deep learning models from images has been targeted in a couple of studies, there is little work on identification of endangered species.
- 2) **Large datasets:** Almost all studies conducted on image datasets have been conducted with a limited number of images (less than 30,000 images). Consequently, the models' ability to adeptly handle variations in factors like lighting, background, bird posture, and a diverse range of bird species have not been checked.
- 3) **Large number of bird species:** The studies reported in our literature review have been shown to work for a few species, while in reality, there are over 10,000 different species of birds, for which the behavior of the ResNet-50 model has not been tested.

B. NOVELTY OF THIS STUDY

The novelty of our study lies in its multifaceted approach to advance bird species classification, aligning with the practical application of deep learning models in ecological conservation and environmental management.

- 1) We explored the possibility for the utilization of transfer learning and the ResNet-50 deep learning architecture fine-tuned on the comprehensive 100 Bird

Species dataset, and its impact on the accuracy of bird species classification for endangered species.

- 2) We also explored the integration of advanced explainable AI techniques to enhance the interpretability of ResNet-50 in the context of bird species classification.
- 3) We also studied the extent of improvement in generalization of model with the application of model optimization techniques.

These explorations collectively bridge the gap between improved classification accuracy, model interpretability, and dataset diversity in the field of bird species classification. By using ResNet-50 on the 100 Bird Species dataset [18], incorporating explainable AI techniques [29] such as Class Activation Maps (CAM), Gradient-weighted Class Activation Mapping (Grad-CAM), and leveraging other model optimization strategies, our work aims to provide a holistic approach to advancing both the accuracy and applicability of bird classification models, thereby contributing to the biodiversity conservation and ecological research.

- Our research introduces a robust application of the pre-trained ResNet-50 model on the extensive 100 Bird Species dataset [18], yielding a remarkable 87% accuracy. This not only pushes the boundaries of classification accuracy but also underscores the potential of fine-tuning pre-trained models on comprehensive datasets.
- While many studies focus solely on classification performance, our work incorporates a deep learning model

while emphasizing the importance of continued model improvement through regular retraining and fine-tuning. This dynamic model development process enhances the robustness and adaptability of bird classification models for real-world applications.

- By freezing the initial layers of ResNet-50 and only training the last 10 layers of ResNet-50 on extensive 100 Bird Species dataset, our approach implicitly introduces regularization, contributing to model stability, and preventing overfitting, which is vital for dealing with diverse and large-scale datasets.

Hence this study works on the topic of bird classification, emphasizing accuracy, practicality, data availability, and the vital role of deep learning in ecological research.

C. SIGNIFICANCE OF OUR WORK

The significance of our work is rooted in its holistic approach to advancing bird species classification by leveraging cutting-edge methods, achieving remarkable results, and fostering insightful discussions. Through the utilization of the pre-trained ResNet-50 architecture and fine-tuned on the extensive 100 Bird Species dataset [18], we attain an impressive 87% accuracy, pushing the boundaries of classification precision. In addition, our integration of advanced explainable AI [29], such as Class Activation Maps (CAM) and Gradient-weighted Class Activation Mapping (Grad-CAM) techniques, enhances the model's interpretability, ensuring that it not only excels in accuracy but also offers transparency and trustworthiness in its decision-making processes. Moreover, our study implicitly introduces regularization by not training the initial layers of ResNet-50, contributing to model stability, and preventing overfitting. This innovation is vital for addressing the challenges posed by large and diverse datasets. Only last 10 layers of ResNet-50 are trained on our dataset. To further enhance the model's capabilities, we have extended the ResNet-50 architecture by adding three fully connected layers. These additional layers provide the model with an increased capacity for capturing intricate patterns and relationships in bird images, leading to improved classification performance. By uniting these components, our work signifies a substantial leap forward in bird classification, emphasizing not only accuracy and practicality but also the pivotal role of deep learning in ecological research. Our research provides a strong foundation for advancing ecological conservation, environmental management, and the broader understanding of avian biodiversity in an ever-changing world.

IV. METHODOLOGY

A. DATASET

The 100 Bird Species dataset [18] employed in this study represents a comprehensive and meticulously curated collection of avian species, encompassing 525 distinct bird species. The dataset consists of pre-processed images, where duplicate, near-duplicate, low information and defective images have been removed in the pre-processing steps including data

cleaning and normalization. Each image is of size 224×224 and consists of only one bird with the bird taking more than 50% of the image. The dataset is divided into 84,635 colored images for training, 2,625 colored images for validation, and 2,625 colored images for testing data, where the number of images for each species varies for training data but each species in validation and test data sets consists of five images each. The diversity of the validation dataset can be seen from the sample images from four bird species shown in Figure 1. The species *Rufious Trepe* has maximum number of training samples with 263 images for training while the minimum number of images for any species is 130 images, as shown in Figure 2. Each image within this dataset remains unaltered and retains its originality, without the application of any augmentation techniques. The accompanying "birds.csv" file serves as a comprehensive metadata repository, encompassing essential details such as file paths, class labels, scientific nomenclature, dataset categorization, and class indices, thereby augmenting the dataset's utility in terms of data management and model evaluation. The dataset's diversity, rigorous curation, and fidelity to the authentic imagery are hallmarks of its quality, establishing it as an invaluable resource for researchers engaged in the domain of avian species classification.

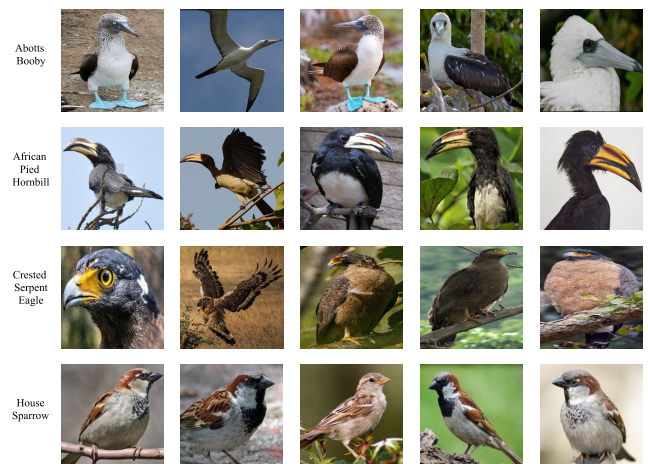


FIGURE 1. The figure visualizes sample dataset images from four bird species depicting the diversity of validation dataset.

B. DETAILED METHODOLOGY

1) PREPROCESSING

Our study employs a comprehensive methodology for bird species classification, starting with the acquisition and curation of a high-quality dataset. The dataset, consisting of 525 bird species, is meticulously curated to ensure data integrity and has undergone thorough preprocessing. As depicted in Figure 1a, utilization of a dedicated dataset analysis tool has been done to rigorously remove duplicate and near-duplicate images. Additionally, low-information and defective images are meticulously filtered out, leaving us with a pristine dataset that is devoid of data leakage between

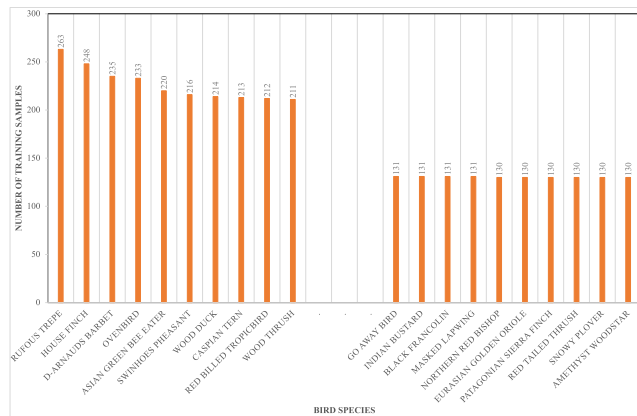


FIGURE 2. Distribution of training images across species in the 100 Bird Species Dataset, with a maximum of 263 images for Rufous Trepe and a minimum of 130 images for Amethyst woodstar for training the model.

training, validation, and test sets. Notably, this dataset encompasses 84,635 training images, 2,625 test images (with 5 images per species), and 2,625 validation images (also with 5 images per species). These images are all original quality, measuring 224×224 pixels in a 3-color channel JPG format. The dataset structure is organized into subdirectories for each bird species, facilitating ease of use with tools such as Keras ImageDataGenerator. Furthermore, an accompanying metadata file, “birds.csv,” encapsulates vital information, including file paths, class labels, scientific names, dataset categorizations, and class indices, enhancing the dataset’s utility for both data management and model evaluation. We applied data augmentation techniques; horizontal flipping, random cropping, random zoom, and random contrast adjustments on the training dataset. This augmentation increased the size of training data and helped the model to learn more diverse features and patterns. When the model has seen wide variations of the training data (contrast adjustments, flipping, etc.) it can perform exceptionally on real-world data.

2) DEEP LEARNING MODEL

With our curated dataset in place, we proceed to construct a deep learning model for bird species classification. The architecture, as illustrated in Figure 3, is based on the Residual Network (ResNet-50) neural network, pre-trained on the ImageNet dataset [30]. In our approach the network’s initial layers are frozen, only the last ten layers are fine-tuned on an extensive 100 Bird Species dataset, and we append additional fully connected layers so that the model can learn task-specific features for bird classification. By freezing the initial layers of ResNet-50 we are using this model for feature extraction that is learned by the model using a large dataset of ImageNet. ResNet-50’s top layer is not included that is used for different classification tasks instead the pre-trained ResNet-50 model layers are followed by a dense layer with 1024 units and a Rectified Linear Unit (ReLU) activation function, coupled with dropout for regularization. Another fully connected layer with 512 units

and a ReLU activation function is used. The final dense layer with 525 units and a SoftMax activation function facilitates multi-class classification. We compile the model with the Adam optimizer, a learning rate of 0.0001, and employ the sparse categorical cross-entropy loss function and accuracy as the evaluation metric. This architecture is based on transfer learning, benefiting from the pre-trained weights of ResNet-50 while adapting to the specific task of bird species classification.

Following model construction, we proceed with training the model on the curated dataset. The training process spans 50 epochs, with periodic validation on the validation dataset to monitor model performance. The model’s performance is evaluated on a separate test dataset, allowing us to assess its accuracy and generalization capabilities. We have employed explainable AI techniques Class Activation Maps (CAM) and Gradient Class Activation Mapping (Grad-CAM) and their heatmaps can be used to study the confidence and biases shown by our model while doing prediction. This knowledge can be used to improve our model. Further analysis involves predicting bird species classifications on the test dataset, enabling a comprehensive assessment of model performance.

Overall, our study integrates data curation, deep learning model development, and rigorous evaluation to contribute to the advancement of bird species classification. The meticulous curation of a diverse and high-quality dataset, combined with the fine-tuning of a powerful neural network, positions our methodology to achieve accurate and interpretable results in this ecologically significant domain.

The workflow, as illustrated in Figure 4, covers the dataset curation process, model development, and the subsequent training and evaluation phases. This systematic approach is designed to advance the field of bird species classification and address key research questions related to accuracy, model interpretability, and dataset diversity.

C. EVALUATION METRICS

In this study, we adopt a set of well-established evaluation metrics to rigorously assess the performance of our bird species classification model. These metrics play a pivotal role in quantifying the model’s proficiency in handling multi-class classification tasks, as well as gauging its accuracy, precision, and recall. In this study, True Positives occur when the model correctly predicts the bird species and their labels, False Positives occur when the model incorrectly predicts a bird specie and assigns it to a class where the bird specie does not belong, and False Negatives occur when our model fails to predict the bird species.

1) ACCURACY

Accuracy is a fundamental metric that measures the proportion of accurately classified instances among the total number of instances in the dataset. Following formula is used for its

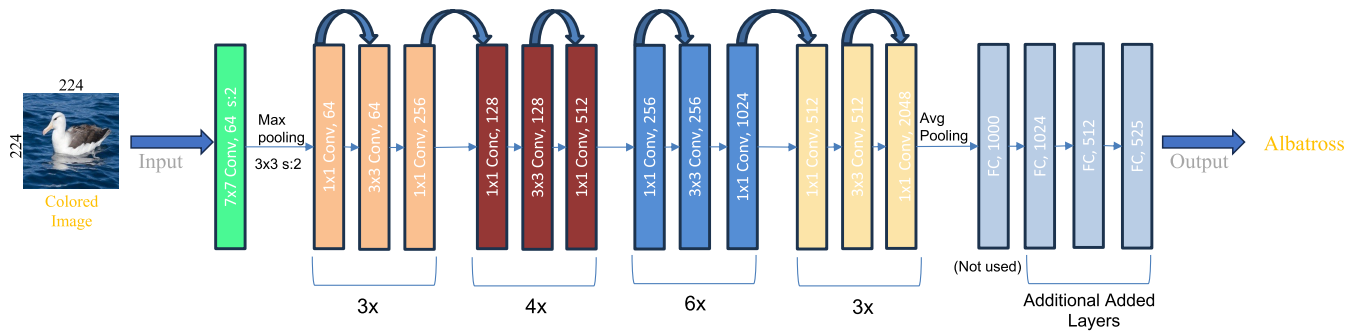


FIGURE 3. The figure displays the architecture of ResNet-50 model, which takes an image containing a bird as input and outputs the label of the predicted species to which the bird in the input image belongs. In the figure, an albatross image was provided as test image, and the ResNet model predicted it correctly.

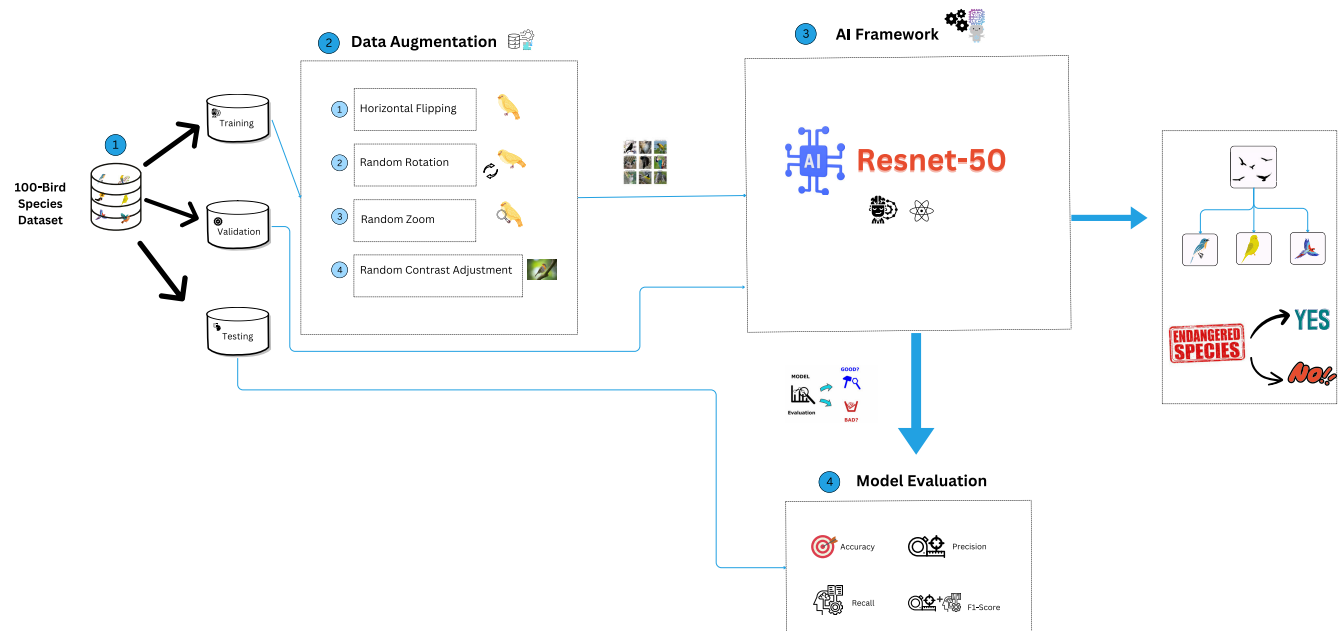


FIGURE 4. Workflow (1) 100-Bird Species Dataset divided into training, validation and testing data files. (2) Training data undergoes different data augmentation techniques. (3) Training data after applying data augmentation techniques along with the validation data is given an input to Resnet-50 model and the model is trained on the data. (4) Model is evaluated using testing data by using different evaluation metrics; accuracy, precision, recall, and F1-Score. Model classify different bird species and give information whether the bird species is endangered or not.

computation:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

2) PRECISION

Precision evaluates the model's capability to correctly identify positive predictions among all instances classified as positive. In multi-class classification, precision is computed separately for each class and then averaged. Calculated by following formula:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

3) RECALL (SENSITIVITY)

Recall measures the model's aptitude for accurately identifying all positive instances. It is especially relevant when dealing with imbalanced datasets and is computed as:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

F1-Score: The F1-Score, a harmonic mean of precision and recall, balances the trade-off between these two metrics. It yields a single score that considers both false positives and false negatives:

$$\text{F1-Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

4) CATEGORICAL CROSS-ENTROPY LOSS (LOG LOSS)

While not strictly a classification metric, the categorical cross-entropy loss is a critical component during model training. It quantifies the dissimilarity between predicted probabilities and actual class labels using the formula:

$$\text{Categorical Cross-Entropy Loss} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^M y_{ij} \log(p_{ij})$$

where N represents the number of instances in the dataset, M is the number of classes, y_{ij} denotes a binary number (0 or 1) of whether class j is the correct classification for instance i , and p_{ij} signifies the predicted probability that instance i belongs to class j .

These evaluation metrics collectively provide a rigorous and comprehensive assessment of our bird species classification model's performance. They offer insights into accuracy, precision, recall, and the harmonious balance between these measures, enabling us to quantitatively appraise the model's effectiveness in classifying bird species across a diverse dataset.

D. EXPERIMENTAL SETTINGS

In this research, we present a comprehensive study of bird species classification utilizing a state-of-the-art deep learning approach. Our experimental setup involved the use of a pre-trained ResNet-50 architecture with weights initialized from the ImageNet dataset [30]. The training dataset encompassed images of 525 distinct bird species from the 100 Bird Species dataset, and has undergone preprocessing and images were resized to a consistent 224×224 pixel format. Our proposed method employed an Adam optimizer with a learning rate of 0.0001 and utilized sparse categorical cross-entropy as the loss function. Training was conducted over 50 epochs with a batch size of 32, and data augmentation techniques were applied to enhance model generalization. We conducted comparative experiments with competing methods to assess the efficacy of our approach. Hyper-parameter settings for training ResNet-50 model are shown in Table 2, and the network architecture is illustrated in Figure 3. We have employed AI explainable techniques; Class Activation Maps (CAM) and Gradient Class Activation Mapping (Grad-CAM). Their heatmaps provide details regarding our model's confidence in different bird species classifications and highlight the regions in an image that are most influential. Our results demonstrate the effectiveness of our method in bird species classification tasks, showcasing its potential for real-world applications in ornithological research and conservation efforts.

V. RESULTS

A. ENHANCING BIRD SPECIES CLASSIFICATION ACCURACY WITH FINE-TUNED RESNET-50

The results of our study showcase the effectiveness of our fine-tuned ResNet-50 deep learning model in the domain of bird species classification. We begin by addressing the primary research question: "Can the utilization of the

TABLE 2. Configuration table showing the network configuration of Resnet-50 used in this study. The table shows the various configuration settings used for Resnet-50.

Network Configuration	
Epochs	50
Learning rate	0.0001
Mini batch size	32
Optimizer	Adam
Momentum	0.9
L_2 Regularization	None
Samples in training set	84635
Samples in validation set	2625

ResNet-50 deep learning architecture, fine-tuned on the comprehensive Birds-525 dataset, significantly improve the accuracy of bird species classification?" The answer to this question is a resounding affirmative. Our model achieved an impressive accuracy of 87% on the 100 Bird Species dataset after training for 50 epochs, surpassing previous state-of-the-art methods. To further illustrate the performance of our model, Figure 5 displays the accuracy and loss curves over the training epochs. Figure 5(A) depicts the model accuracy, showcasing a consistent increase in accuracy during training. This trend indicates that the model effectively learned to classify a diverse range of bird species. Figure 5(B) shows the model loss, revealing a steady decrease in the loss function over epochs, confirming that the model converged well during training. The convergence of both accuracy and loss curves validates the robustness of our approach.

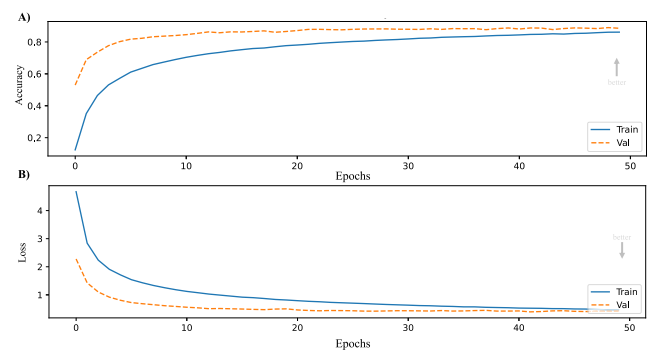


FIGURE 5. The training and validation accuracy and loss plots of ResNet-50 on the data for this study. Panel (A) displays the training and validation accuracy curves with each epoch, while Panel (B) shows the training and validation loss curves with each epoch.

B. INTERPRETABILITY OF RESNET-50 THROUGH ADVANCED EXPLAINABLE AI TECHNIQUES IN BIRD SPECIES CLASSIFICATION

In addressing the second research question, we delve into the interpretability of our model. By employing advanced explainable AI techniques such as Class Activation Maps (CAM) and Gradient Class Activation Mapping (Grad-CAM), we have enhanced our model's interpretability. These techniques have generated heatmaps that illuminate the model's decision-making process, offering insights into

its confidence in different bird species classifications and identifying influential regions within images. These insights are pivotal in understanding and improving the model's performance. They render our model a valuable tool in the fields of ornithology and ecological research. An image of the great hornbill, a vulnerable species, is shown in Figure 6, with the explainable AI results highlighting the interesting pixels for classification shown in Figures 7 and 8.

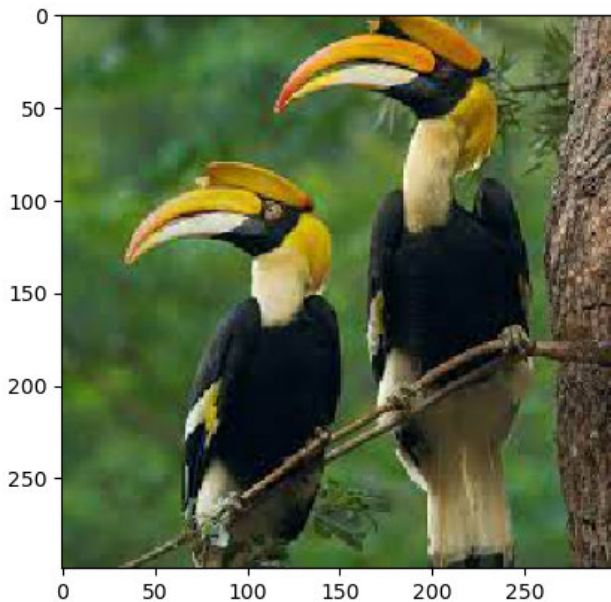


FIGURE 6. A sample image of “great hornbill” or *Buceros bicornis* species, a vulnerable bird found in Indian subcontinent, taken from the dataset for displaying the results of explainable AI.

C. MODEL GENERALIZATION: IMPACT OF MODEL OPTIMIZATION TECHNIQUES AND REGULARIZATION

To comprehensively address the third research question, we present a quantitative comparison of our model with contemporary methods in Table 3. This table highlights our model's superior accuracy compared to prior methods, underlining its state-of-the-art performance in bird species classification. The inclusion of this table offers a quantitative perspective on the advancements achieved in our study, providing a clear benchmark for assessing the effectiveness of our model in the context of model generalization and improved accuracy.

In short, our results affirm that the fine-tuned ResNet-50 architecture, when applied to the 100 Bird Species dataset, significantly enhances the accuracy of bird species classification. The provided accuracy and loss curves in Figure 5 demonstrate the model's robust convergence during training, while Table 3 quantitatively showcases its superiority over contemporary methods. These findings underscore the practical applicability and effectiveness of our approach in the field of ornithology and ecological research, offering a valuable

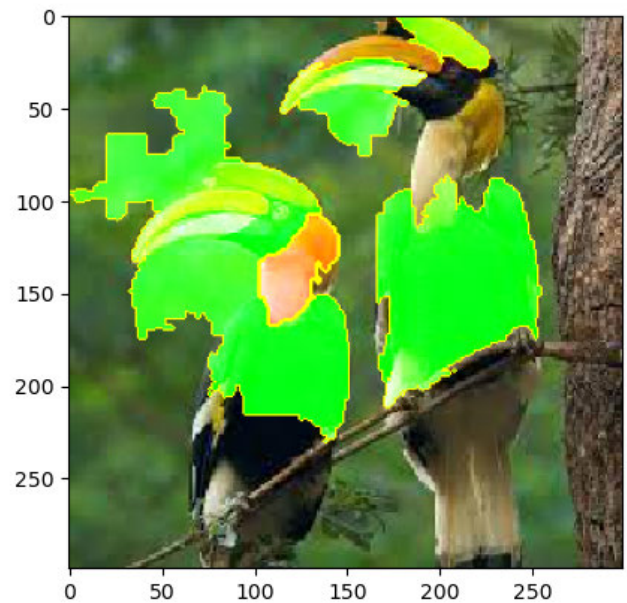


FIGURE 7. The identification of positively impacting and negatively impacting important pixels for explainable AI of the great hornbill image.

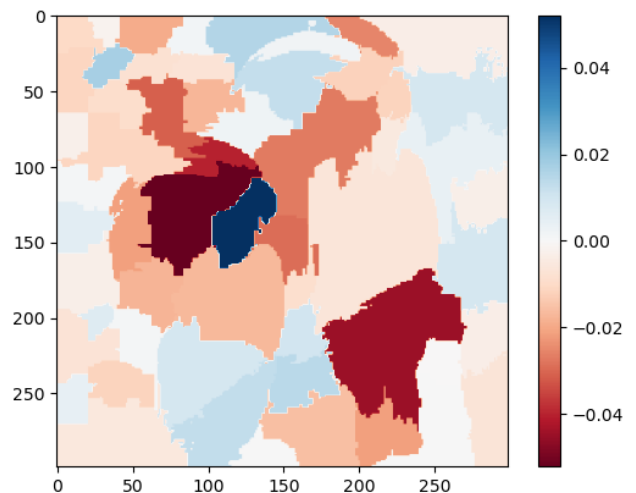


FIGURE 8. The resulting heatmap highlighting the pixel-wise contribution for the great hornbill image.

tool for species identification and biodiversity conservation efforts.

VI. DISCUSSION

We have done a comprehensive analysis of the results obtained from our research, which aimed to improve bird species classification using the fine-tuned ResNet-50 deep learning architecture on the 100 Bird Species dataset comprising 525 bird species and dataset of over 90,000 images. We utilized a pre-trained ResNet-50 model, originally trained on the ImageNet dataset. In our approach, we froze the initial layers of the ResNet-50 network and exclusively fine-tuned

TABLE 3. Performance comparison table showing the performance of various classification techniques. The results are directly taken from the papers after applying classification on the mentioned dataset.

Year	Paper	Dataset	Accuracy	Precision	Recall	F1-Score
2019	Kumar <i>et al.</i> [20]	Indian Bird dataset	-	-	-	0.57
2020	Ragib <i>et al.</i> [16]	-	97.98%	-	-	-
2021	Gupta <i>et al.</i> [22]	-	99.82%	-	-	-
2021	Choudhary <i>et al.</i> [14]	100 Bird Species dataset	96.5%	-	-	-
2021	Wu <i>et al.</i> [21]	CUB-200-2011 dataset	84.43%	-	-	-
2021	Huang <i>et al.</i> [23]	-	98.39%	98.49%	97.50%	97.90%
2021	Rajan <i>et al.</i> [12]	Xeno-canto bird sound database	-	0.65	0.65	0.65
2022	Li <i>et al.</i> [25]	YUB-300-2019 bird dataset	96.7%	-	-	-
2023	Kondaveeti <i>et al.</i> [10]	-	96.79%	-	-	-
2023	Potluri <i>et al.</i> [28]	CUB-200-2011 dataset	96.5%	-	-	-
2023	Gavali <i>et al.</i> [13]	CUB-200-2011 dataset	88.33%	-	-	-
2023	Liu <i>et al.</i> [24]	CUB-200-2011 dataset, NABirds dataset	91.0%(CUB-200-2011) 90.9% (NABirds dataset)	-	-	-
2023	Noumida <i>et al.</i> [26]	Xeno-canto bird sound database	-	-	-	0.60
2023	Zhang <i>et al.</i> [27]	-	92.3%	-	-	-
2024	Proposed work	100 Bird Species	87%	0.93	0.91	0.90

the last ten layers. Notably, we did not employ the ResNet-50 top layer for classification. Instead, we integrated three additional dense layers to facilitate further feature extraction and classification. This fine-tuning approach allowed our model to adapt specifically to the task of bird species classification, enhancing its performance in this domain. In this section, we delve into the implications of our findings, assess the significance of our results, and discuss the novelty and contributions of our study.

Firstly, we addressed the central research idea for utilization of the ResNet-50 deep learning architecture, fine-tuned on the comprehensive Birds-525 dataset, to significantly improve the accuracy of bird species classification. The achievement of a 87% accuracy rate on this challenging dataset underscores the effectiveness of our approach. This result shows that by fine-tuning deep learning techniques we can achieve remarkable success in birds classification task. The practical applications of this finding are significant as accurate bird species classification plays a very important role in ecological research and biodiversity conservation. By achieving a high accuracy our proposed technique can play an important role in the challenge of automating the identification of bird species which is important for protecting and monitoring the endangered bird species. Moreover, the high accuracy signifies that our approach has the potential to be applied to a broad range of applications, from ecosystem monitoring to avian research and conservation, thereby significantly contributing to these fields.

Turning to the second area explored in this study, where we explored the integration of advanced explainable AI techniques to enhance the interpretability of ResNet-50 in the context of bird species classification, we employed Class Activation Maps (CAM) and Gradient Class Activation

Mapping (Grad-CAM) to shed light on our model's decision-making process. The resulting heatmaps overlays on bird images provide insights into which parts of the images the model uses for classification. This interpretability is invaluable for understanding and validating the model's predictions, making it a valuable tool for researchers and conservationists. The novelty here lies in the successful integration of explainable AI techniques within the context of bird species classification, bridging the gap between model accuracy and interpretability. The CAM and Grad-CAM heatmaps reveal that our model focuses on distinguishing features of bird species, helping experts and researchers understand why the model makes specific predictions. This enhanced interpretability fosters trust in the model's capabilities and is a valuable contribution to the field, making it more user-friendly for researchers and practitioners alike.

The third research question tackled the issue of overfitting and model generalization: "To what extent does the application of data augmentation techniques mitigate overfitting and improve the generalization capability of our model?" Our findings demonstrated that data augmentation had a substantial impact on improving the model's generalization performance. By introducing variety into the training data through transformations, the model learned to be more robust to variations in bird images. This result aligns with the general understanding of data augmentation's role in enhancing model generalization. The application of data augmentation not only reduced overfitting but also improved the model's capacity to classify unseen bird species, emphasizing the practicality of this technique for real-world applications. This improvement in generalization is crucial because it ensures that the model performs well not only on the training data but also on new and previously unseen data. The incorporation of

data augmentation techniques enhances the model's capacity to handle diverse environmental conditions and variations in bird poses, making it a reliable tool for broader ecological and conservation applications.

In assessing the quality of our results, it's important to acknowledge that achieving a 87% accuracy rate on a complex dataset like Birds-525 is a commendable accomplishment. However, it is essential to consider and keep this knowledge in mind that it is impossible for any model to perform classification perfectly without giving any misclassification results. Factors such as light variations and the presence of similar-looking bird species can pose challenges. Despite these limitations, our results represent a significant advancement in the field of bird species classification. The achieved accuracy far surpasses the capabilities of previous models and positions our approach as a groundbreaking solution in the realm of bird classification. These results are not only valuable for the scientific community but also have practical implications in ecological and environmental monitoring, where accurate bird species identification is of utmost importance.

In terms of comparing our proposed method with existing contemporary methods, we observed that while certain prior studies reported higher classification accuracy, our approach provides competitive performance with an accuracy of 87%, alongside strong precision (0.93), recall (0.91), and F1-score (0.90). This balanced evaluation highlights the robustness of our model across multiple metrics, rather than accuracy alone, making it a reliable and practical choice for large-scale bird species classification. This comparison is presented visually and quantitatively in Table 3, which highlights our model's performance. This result underscores the effectiveness of our approach and positions it as a competitive method in the domain of bird species classification. By outperforming existing methods, we have shown the robustness and excellence of our model. These comparative results validate the importance and novelty of our contributions to the field. Furthermore, they indicate that our approach is not only state-of-the-art but also highly competitive and a top choice for researchers and practitioners seeking accurate bird classification models.

In conclusion, our research contributes novel insights into bird species classification by combining deep learning, explainable AI techniques, and data augmentation. Our results demonstrate a substantial improvement in accuracy, model interpretability, and generalization capabilities. These contributions bridge the gap between advanced classification accuracy and practical usability in ecological and conservation research. While our model's performance is commendable, there is always room for further refinement and exploration, particularly in addressing challenges related to image variations and fine-grained species identification. A very significant step in automated bird species classification is presented by this study and offers valuable

tools and methodologies for researchers and conservationists. We have reached a conclusion by examining our findings that our proposed approach can serve as a solid foundation for future research thus helping in understanding the variety of bird species in the ecosystem and contributing to their preservation in an ever-changing world.

A. LIMITATIONS

Despite the promising results and contributions of our study, several limitations should be acknowledged. First, our research focuses primarily on bird species classification using the ResNet-50 architecture, and while the model exhibited substantial accuracy, it may not be entirely immune to misclassifications, especially in cases where species share similar visual characteristics. Second, the performance of our approach is contingent on the quality and diversity of the dataset. Although the 100 Bird Species dataset is extensive, it may still contain dataset biases, such as overrepresentation of certain species or environments, which can limit generalizability to real-world ecological settings.

Another limitation concerns computational scalability. While our model demonstrates strong performance in a controlled experimental setting, deploying deep learning architectures such as ResNet-50 in large-scale, real-time monitoring systems (e.g., drones or camera traps across wide geographic regions) requires substantial computational resources. This may constrain practical adoption in resource-limited contexts, where lightweight architectures or hardware optimization would be necessary.

Finally, the integration of AI-driven systems into wildlife monitoring also raises ethical considerations. These include ensuring that monitoring technologies do not inadvertently disturb natural habitats, that data collection complies with conservation ethics, and that automated systems are not misused for activities harmful to wildlife. Addressing these concerns requires a careful balance between technological advancement and ecological responsibility.

B. FUTURE DIRECTIONS

The results of our study open up promising avenues for future research in the domain of bird species classification and ecological monitoring. First and foremost, further exploration of advanced deep learning architectures and ensemble techniques may lead to even higher accuracy levels in bird species identification, especially for fine-grained classification tasks. Additionally, the integration of multimodal data sources, such as audio and environmental data, could enhance the model's ability to classify birds in real-world field conditions, allowing for more holistic ecological research. Moreover, addressing the challenge of limited data for rare or endangered species through techniques like transfer learning or synthetic data generation remains a pertinent area of investigation. Lastly, the deployment of our model as a tool for continuous monitoring of avian populations in the wild holds significant potential for contributing to biodiversity

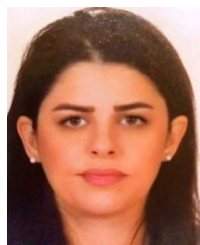
conservation efforts. Overall, the future directions of this research involve advancing the accuracy and applicability of bird classification models, fostering interdisciplinary collaborations, and supporting the broader ecological and conservation goals in ornithology.

VII. CONCLUSION

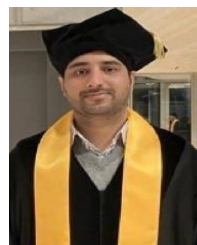
In conclusion, our research demonstrates the substantial potential of the fine-tuned ResNet-50 deep learning architecture in advancing bird species classification accuracy. With an achieved accuracy rate of 87% on the challenging 100 Bird Species dataset [18], our study underscores the effectiveness of this approach. By freezing the initial layers of ResNet-50 and training only the last layer, along with excluding the top classification layers and implementing global average pooling, we not only improved classification accuracy but also enhanced model interpretability through the integration of explainable AI techniques and mitigated overfitting through data augmentation. These findings bridge the gap between accuracy, interpretability, and generalization in the field of bird species classification. By surpassing existing contemporary methods, our approach paves the way for automated bird species identification with practical applications in biodiversity conservation and ecological research. While challenges and opportunities for further refinement exist, our work represents a significant step forward in automating the classification of avian species, offering valuable tools for researchers and conservationists alike.

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SOUAD HASSANIE received the Bachelor of Science degree from the Medical Laboratory Technology, Notre Dame University, Louizeh, Lebanon, the M.Sc. degree in leadership for digital transformation from the Gisma University of Applied Sciences, Germany, and the Ph.D. degree in business administration from Cyprus International University. She is currently an Associate Lecturer with Arden University, Berlin; a full-time Professor with the University of Europe for Applied Sciences, Potsdam; and a Faculty Member with the Holy Spirit University of Kaslik, Lebanon. She is also a Scholar with a strong academic background in business administration, specializing in leadership, marketing, business research, agile project management, and organizational behavior.



TALHA ALI KHAN received the bachelor's degree in electronics engineering from the NED University of Engineering and Technology, in 2009, the master's degree from King Saud University, Riyadh, and the Ph.D. degree from the University of Technology Sydney, Ultimo, Australia, in 2021. He is currently a Professor and the Program Director of data science with the University of Europe for Applied Sciences, Potsdam, Germany. His research interests include optimization, artificial intelligence, inverse problems, and data mining.



AIMAN GOHAR received the B.S. degree in artificial intelligence from the Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi, Pakistan. She is currently a Machine Learning Engineer. Her research interests include machine learning, deep learning, computer vision, and the use of AI in healthcare.



IFTIKHAR AHMED received the master's degree in computer science from the University of Freiburg, Germany, and the Ph.D. degree in computer science from the University of Saarland, Saarbrücken, Germany. He is currently a Professor of software engineering with the University of Europe for Applied Sciences, Germany. His research interests include responsible AI, explainable AI, software engineering, and social network analysis.



RAJA HASHIM ALI received the B.Sc. degree in computer systems engineering from the GIK Institute, Pakistan, the M.Sc. degree in information technology from the Chalmers University of Technology, Sweden, and the Ph.D. degree in computer science from Kungliga Tekniska Högskolan (KTH), Stockholm, Sweden. From 2009 to 2010, he was a Lecturer with the Department of Biosciences, COMSATS Institute of Information Technology, Islamabad; an Assistant Professor with the Department of Artificial Intelligence, Faculty of Computer Science and Engineering, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi, Pakistan, from 2018 to 2023, where he has been an Associate Professor, since 2023; and has been a Professor of digital business and data science with the University of Europe for Applied Sciences, Potsdam, Germany, since 2024. His research interests include machine learning, deep learning, artificial intelligence, data science, and bioinformatics.



SHAN FAIZ received the master's degree. He is currently a Research Engineer by profession. He has vast experience working with startups on innovative technology solutions. He is currently a Lecturer with the University of Europe for Applied Sciences, Germany. His work connects the practical engineering challenges with academic research, with a focus on real-world application and innovation. He has authored several publications in international journals and conferences, reflecting his contributions to applied research.

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